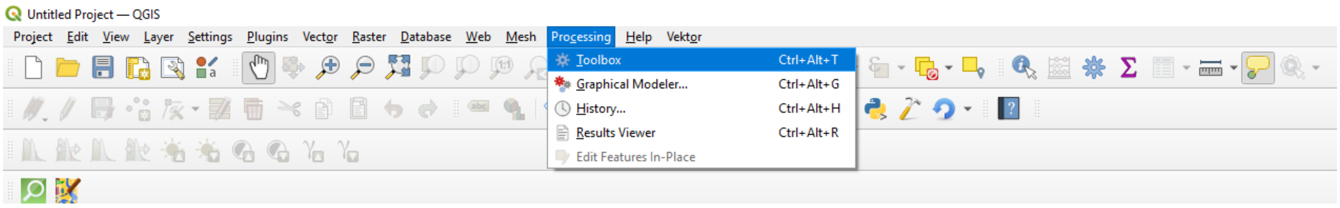
QGIS recipes

Where to learn QGIS?

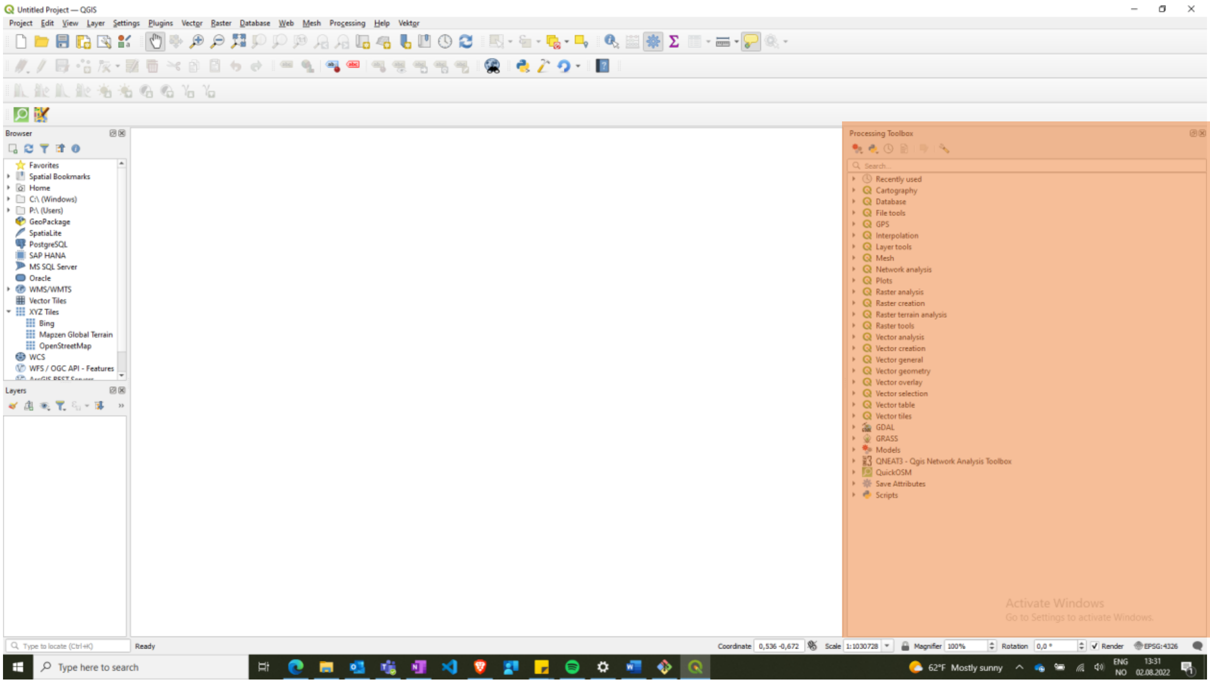
If you want to learn how to use QGIS there are several references you should look at. First of all you have the official QGIS user guide: https://docs.qgis.org/3.22/en/docs/user_manual/index.html. Secondly, the one I used was QGIS Tutorials and tips: https://www.qgistutorials.com/en/#. There were several small follow along projects that give you a sense of what can be done in QGIS. I did the basic GIS operations module and took a quick look at the Python scripting (PYQGIS) module. In general, if you want to work with PYQGIS the official PyQGIS Developer Cookbook is a good place to look: https://docs.qgis.org/3.22/en/docs/pyqgis_developer_cookbook/index.html. Network analysis module can also be interesting from QGIS Tutorials and tips. Also if you work with QGIS or PyQGIS, then the GIS stackexchange(https://gis.stackexchange.com/) is a great place to find help. Most often people have struggled with similar things, so you should find pointers on how to proceed. One thing to keep in mind is that there is a difference between QGIS2 and QGIS3, and especially between the PyQGIS3 and PyQGIS2, as they have different APIs and PyQGIS3 is based on python3, while PyQGIS2 is based on Python 2.

Importing – QGIS model

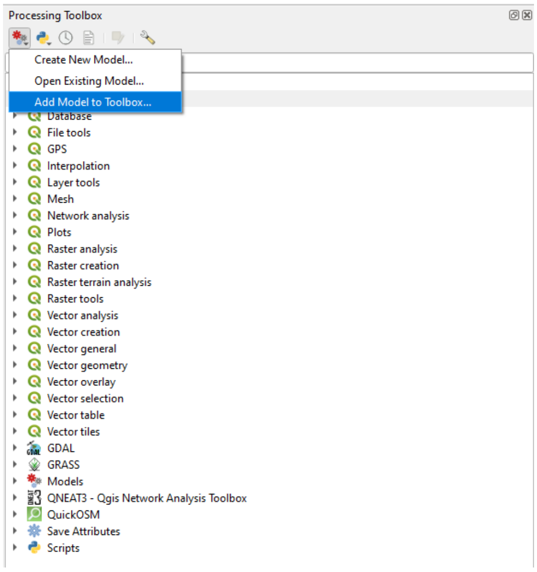
1. Go to processing and click the toolbox.



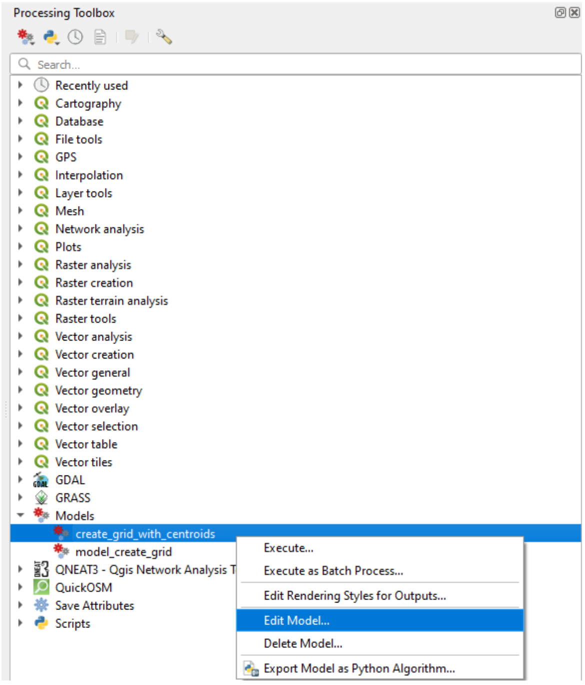
2. The window on the right side of the QGIS interface should appear. I have marked it with an orange color.



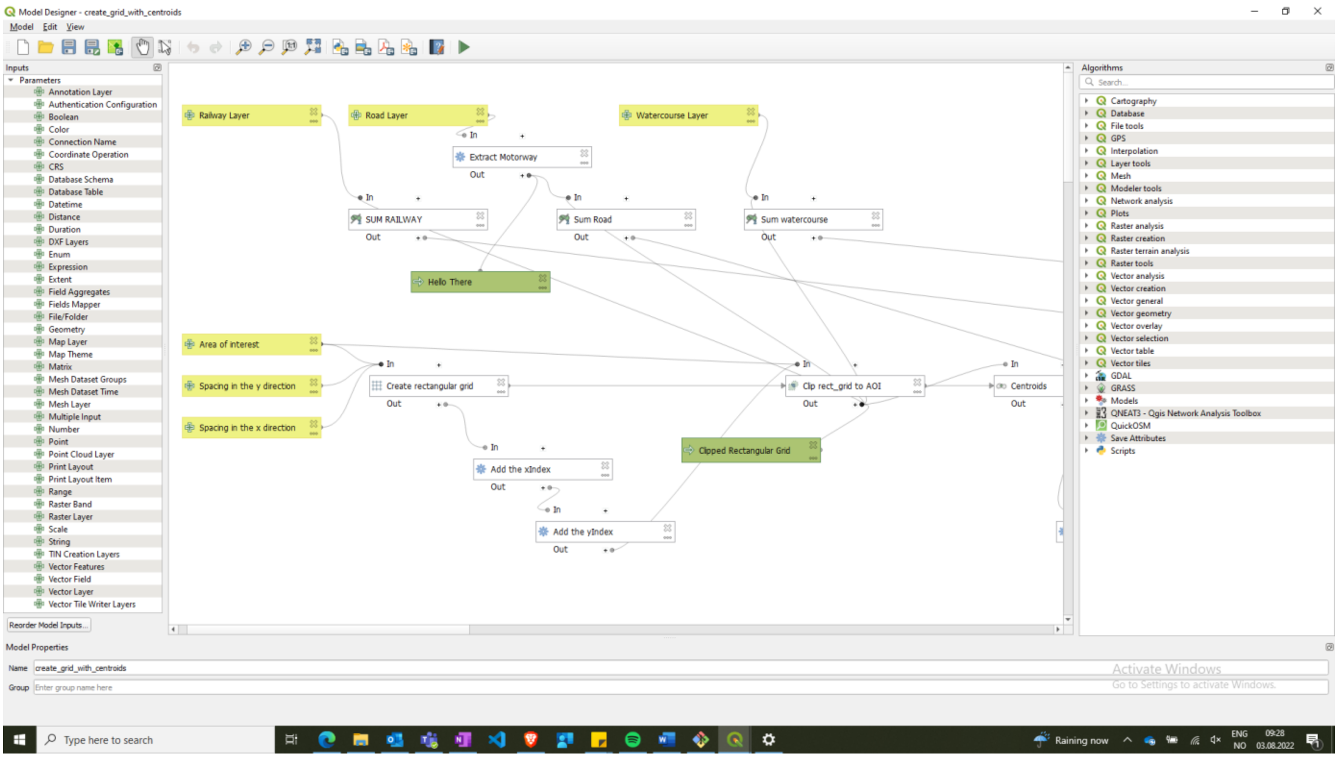
3. Then clicking on the button with the cogwheels and selecting “Add model to toolbox” will allow you to import a model from your .model3 file.



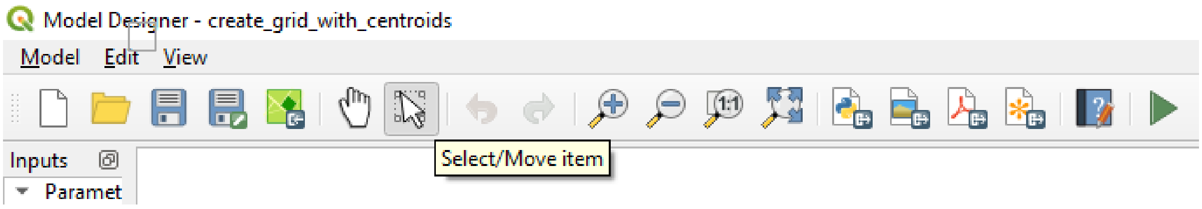
4. Once you have imported the model you can find it under the Models menu. If you want to open the model and look at how it works or want to edit it, just right click the name of the model and click on edit model. Note that there is also an option to export the model as a Python algorithm, if you'd rather work with python scripts.



5. Graphical modeler looks like the following:



On the left side we have a parameters window where you can select what kind of input you want your model to take in. On the right side you find the algorithms panel. These are the same as the algorithms from the processing toolbox. So if you can accomplish your task in QGIS using only the algorithms from the processing toolbox, then you can group them together into a workflow in the graphical modeler and automatize the process. In the middle you have the canvas where you see the logic behind your model. The citrus green boxes denote input, green boxes denote output, and white boxes denote algorithms. The lines between boxes show what each box uses as input and where its input is being used.



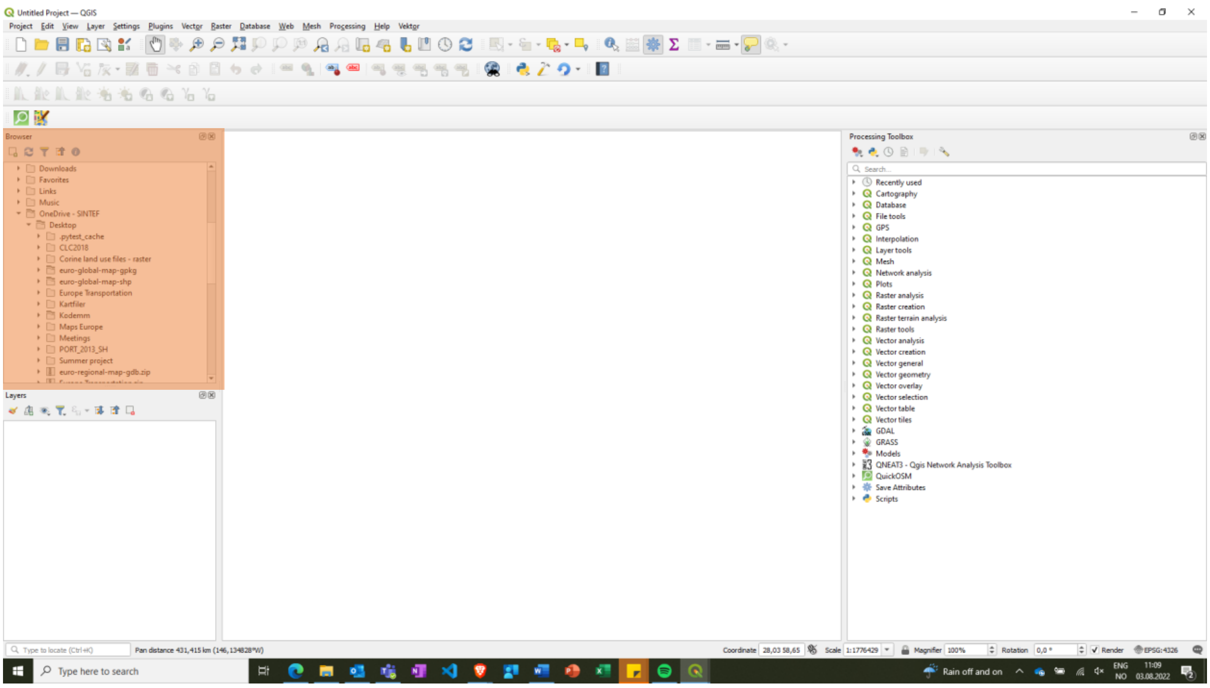
Last tip. If you select “Select/Move item”, then you can double click the boxes and open them. There you can find out what exactly they do. Also, in that popup window there is a section called comments, where I have described what some of the functions do.

How I personally familiarized myself with the graphical modeler was to search “graphical modeler QGIS” and then watch some YouTube videos. That will give you a sense of how to use that functionality of QGIS.

Where we got the data from

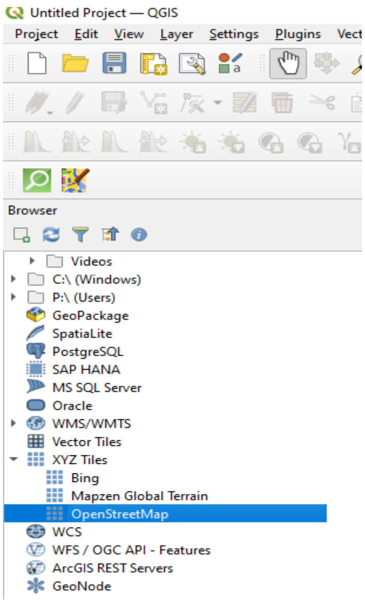
In order to analyze transportation along roads, rails and rivers, we need some information about these network systems. This information was found at EuroGeographics web page(https://eurogeographics.org/). We have used EuroGlobalMap from https://www.mapsforeurope.org/datasets/euro-global-map. You can get the map data sent to your email from https://www.mapsforeurope.org/access-data. You will get the files as Shapefile or GeoPackage. Use whichever one you want. From the data folder we will need three files: RoadL.shp, RailrdL.shp and WatrcrsL.shp. These files are linestring vector layers with the network for road, railroad and rivers, respectively. For more information about these layers and what attributes they possess go to the documents folder and explore the data specification file.

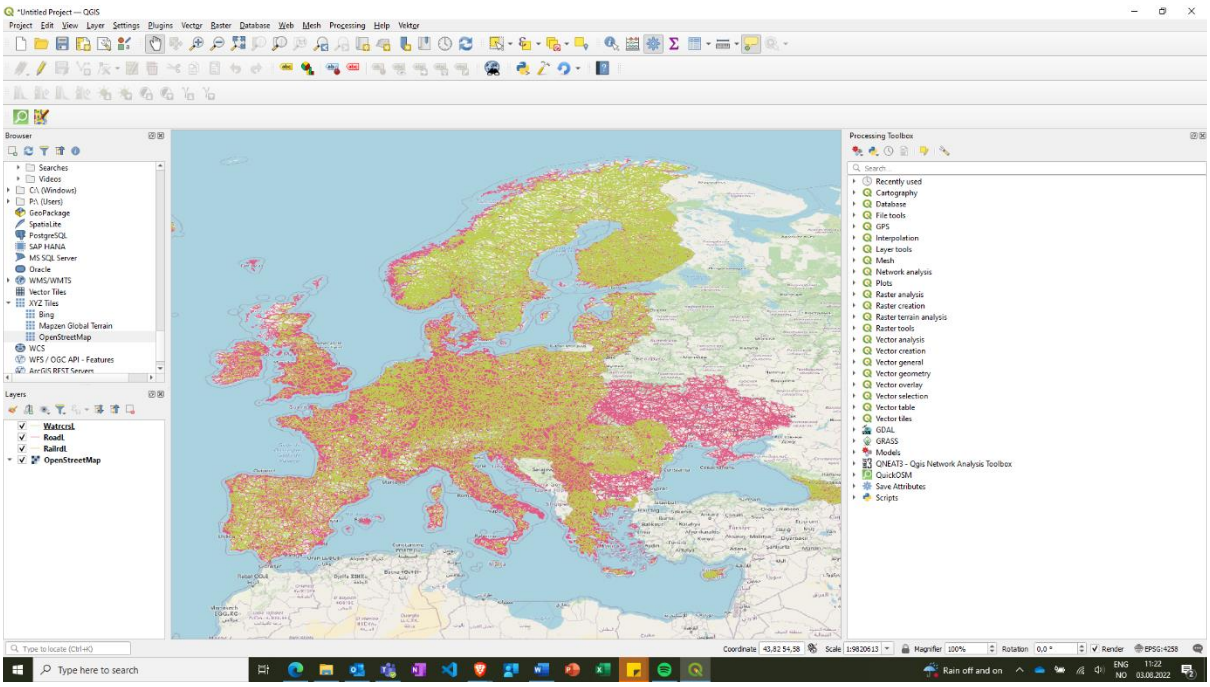
In QGIS you can just use the browser window marked in orange to navigate to the EuroGlobalMap directory and select the required layers.



How to apply the QGIS model

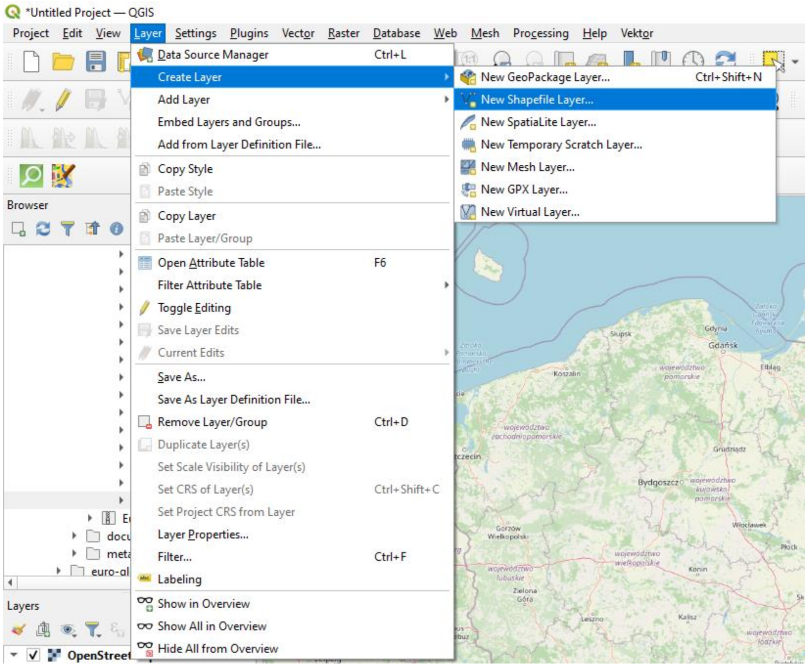
1. Beginning from a blank project. Go to the Browser window on the left side. Then find XYZ Tiles and drag OpenStreetMap onto the map canvas in the middle.
2. Import the Roadl.shp, Railrdl.shp and watrcrsl.shp from the EuroGlobalMap as described above. Now your map should look somewhat like mine does below.



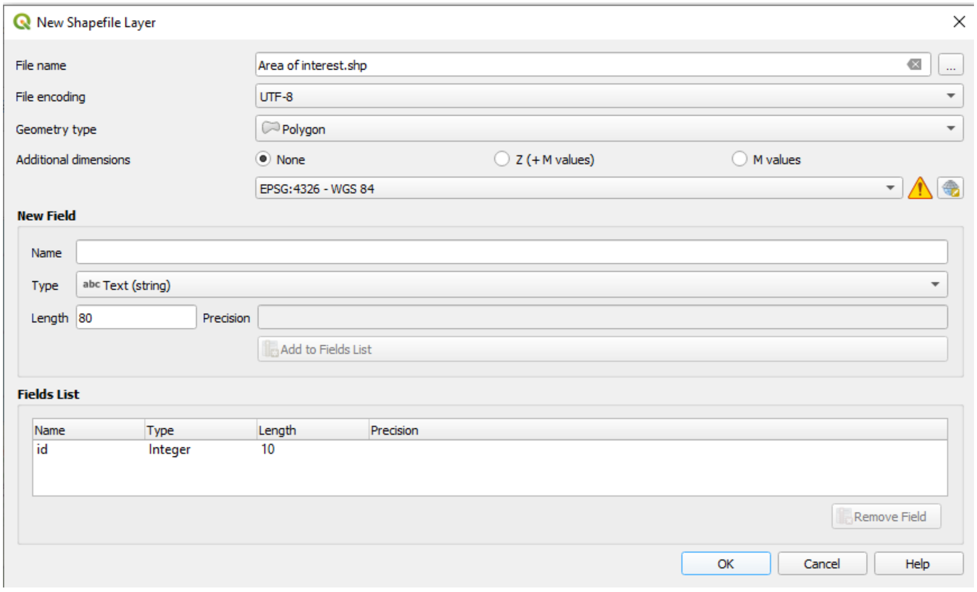


3. You may now uncheck the layers RoadL, RailrdL and WatrcrsL in the layers window in the bottom left corner as they are obstructing our view of the map. Alternatively, you can drag OpenStreetMap layer above all other layers to hide the other layers behind it.

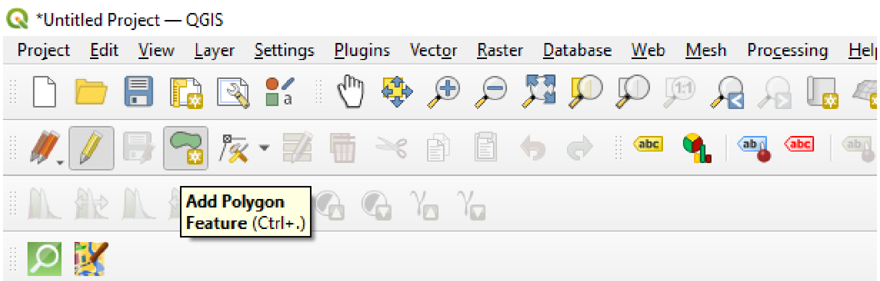
4. Now we may create a new shapefile layer. Go to Layers, create layer and choose new shapefile layer.



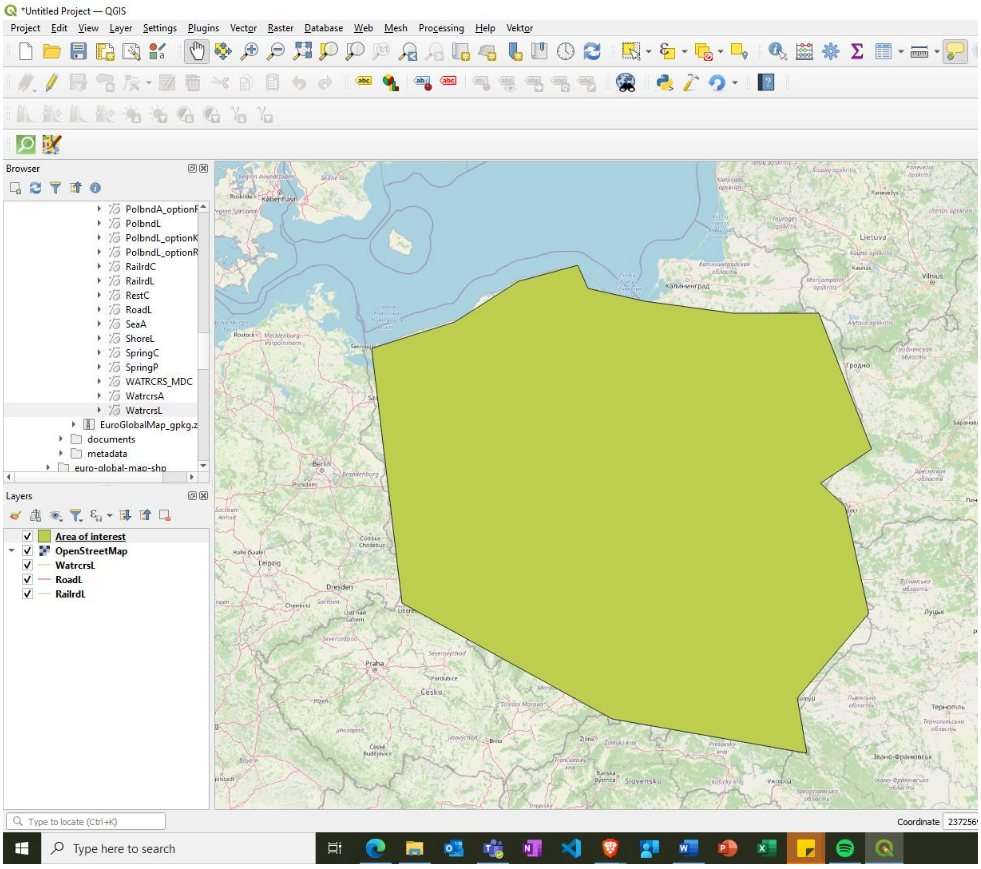
5. Give the new layer a name and choose Geometry type to Polygon and click ok. The projection doesn't really matter here.



6. Now click on the Toggle editing button. the yellow pen button in the top left corner. Thereafter click on the add Polygon feature, the green area button.

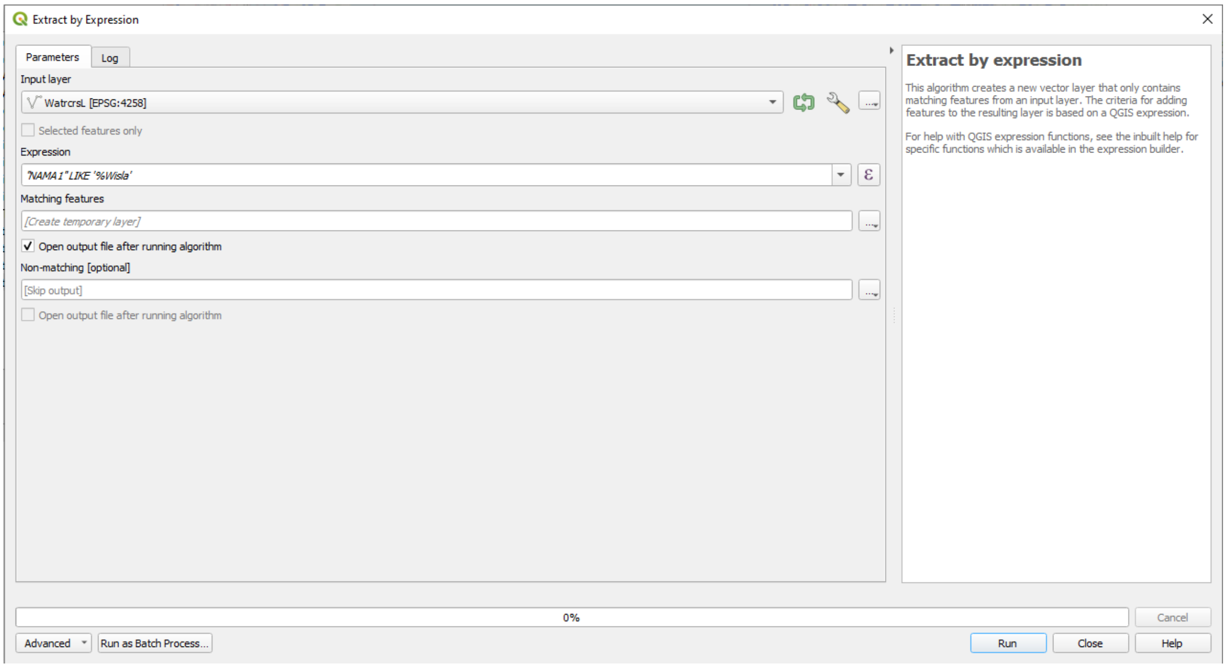


7. Now left click the map canvas to define your area of interest. Once you are done, right click your map to finalize the polygon. Just give it some id number and click ok. If you are unhappy or want to modify your polygon, just click the Vertex tool button and you can edit your polygon. This is the button right beside the add polygon feature. Once you are done you can also unclick toggle editing. The result should look something like this

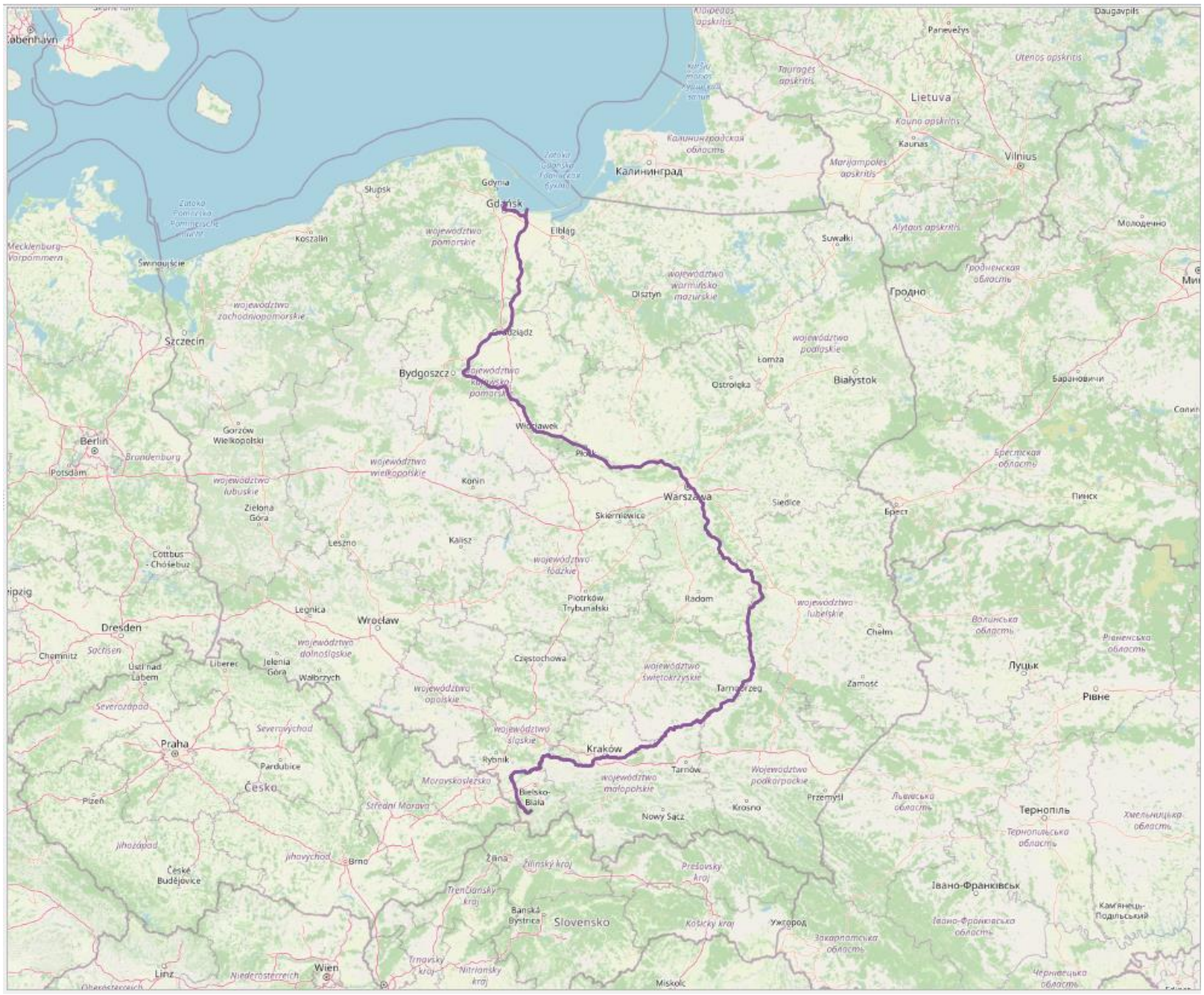


Tip: If you double click on the layer in the layers window in the bottom left corner and go to symbology, you can play around with the styling of your layer, and also change the opacity. Lowering the opacity is good to be able to see the features in the layers below the one you are editing.

8. In this case we want to analyze Poland and the river we want to transport along is called Wisla. We want to extract this river from WatrcrsL layer. And we can do this as the name of the rivers are stored in the layer. Therefore, go to the processing toolbox and find the extract by expression algorithm. Select the WatrcrsL layer and write the following expression in the expression box: “NAMA1” LIKE ‘%Wisla’. Then you can click run.



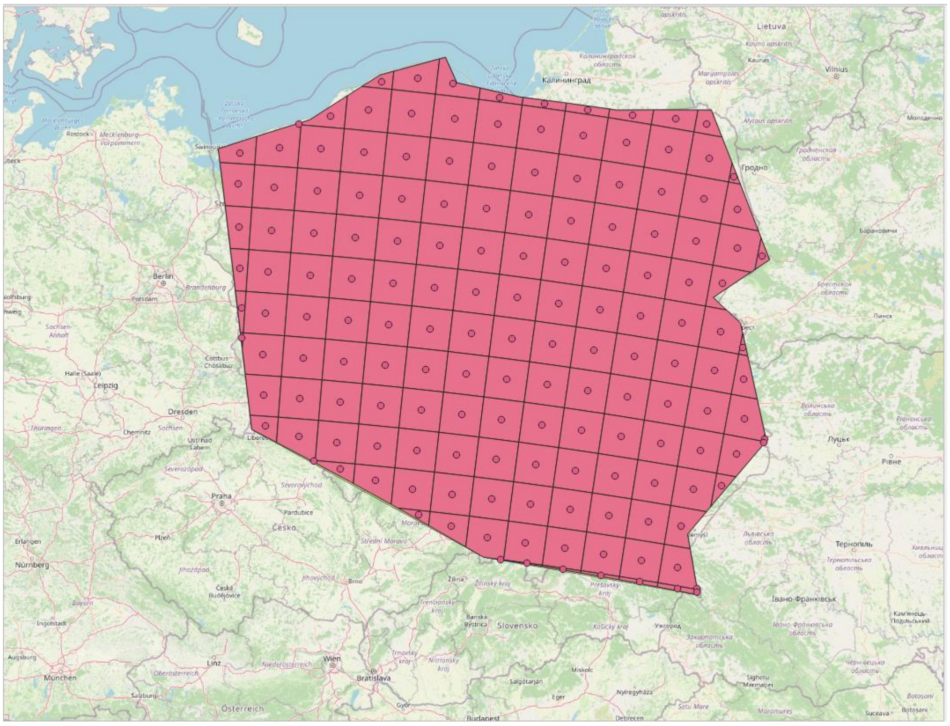
9. You should get a layer like the one shown below. The layer has the name Matching Features in the bottom left corner. If you want to change that, you can right click the layer and choose the rename the layer to give a more meaningful name.



10. Now go to the processing toolbox, find and open the models subdirectory and you should find your imported model “create_grid_with_centroids”. Double click it. Once the input

window pops up. Fill in the Area of Interest with your Area of Interest layer. Then define the size of your grid in m. I have chosen 50km in this example. Fill in the transportation layers as below. If you want to save the information about the gridcells in a CSV files then click on the dots and select where you want to save the file. If you want to save the Clipped Quadratic Grid and the finalized centroids, then you can do it as well by clicking on the dots on the right side and selecting where to save it. If not, then these will be saved as temporary layers that will disappear once you close the project/QGIS. Now click run.

11. You should get a beautiful looking grid with points. Just like the one below.



Producing grids with different size and running a shortest distance algorithm on QGIS